

Financial Literacy Confidence and Retirement Planning: Evidence from China

Abstract

Though there is ample empirical evidence demonstrating the relationship between objective financial literacy and retirement planning, we have limited understanding on the role of subjective financial literacy and financial literacy confidence level in individual's retirement planning. In this paper, we conduct a survey to examine how individual's financial literacy confidence bias affects their retirement planning behaviors in China. The survey measures both the respondent's subjective and objective financial literacy, as well as their assessment and actual actions with respect to their retirement planning. We find that financial literacy overconfidence (under-confidence) significantly promotes (demotes) individual's retirement planning behaviors. Our results contribute to understanding the effect of subjective financial literacy on retirement planning.

Keywords: Financial literacy; Subjective financial literacy; Objective financial literacy; Overconfidence; Retirement Planning

1. Introduction

Financial well-being in retirement is important as the individuals are taking an increasing responsibility for their retirement nowadays. Many researchers have devoted to studying why individuals do not plan and save enough for retirement, and have found an important relationship between financial literacy and retirement planning behaviors. Previous works have demonstrated that those who are less financially literate prepare less for their retirement and amass less retirement wealth in many countries (Behrman et al., 2012; Lusardi and Mitchell, 2011, 2011b; Lusardi, 2008).

More recently, some literature has found mismatches between actual and self-assessed financial knowledge (Lusardi and Mitchell, 2011a). It is critical to understand comprehensively how subjective financial literacy affects retirement planning, in addition to the effect of actual financial literacy. Empirical evidence has revealed that individual's optimistic bias towards their financial literacy affects their financial decision behaviors, including retirement planning, household saving and investment behaviors (Van Rooij et al, 2012; T. Xia et al, 2014).

However, we have limited understanding on whether and how individual's subjective financial literacy and financial literacy confidence level affect their retirement planning, especially on their actual preparation actions. In the limited existing works, the measurement of retirement planning adopted is on the basis of the respondent's self-assessment to some extent. For instance, a common question measuring the retirement preparedness is to ask the respondents to assess how much or how well they have thought about retirement. A potential concern may arise when the overconfident groups overstate their subjective financial literacy and retirement preparedness simultaneously. Thus, it is meaningful to confirm the effect of financial literacy bias on not only the individual's retirement planning but also the actual actions they undertake to prepare for retirement, like purchasing a variable annuity.

To that end, this paper investigates whether the financial literacy overconfidence or under-confidence bias affects individual's retirement planning and actions with evidence from China. Financial literacy overconfidence is measured by an indicator of the difference between an individual's subjective and objective financial literacy scores. In addition to the questions on the respondent's preparedness and awareness of retirement planning, our survey also measures actual preparations by asking whether they have participated in private pension and purchased a variable annuity.

The main contribution of this paper is that we find the effect of financial literacy confidence on individual's retirement planning and preparation conducting with data of China. Our empirical results suggest that respondents with more confidence in their financial knowledge have a higher propensity to plan and take actual actions for retirement. Moreover, we also find that individuals who are under-confident about their financial literacy are less likely to plan for retirement well. This finding contributes to the growing literature on

financial literacy around the world and its effect on individual's retirement planning.

The second contribution is that our database allows us to be the pioneer to have a glance at the financial literacy and retirement planning in China. We show that a large proportion of respondents in our sample display a lack of financial knowledge, but tend to be optimistic about their financial knowledge. Our empirical result also supports the widely-acknowledged finding that the actual financial literacy contributes to one's retirement planning.

The remainder of the paper proceeds as follows. Section 2 gives a brief literature review on financial literacy, overconfidence, and its effects. Section 3 presents the data and introduces the measurement for financial literacy and retirement planning in the survey. Section 4 analyzes the relationship between financial literacy overconfidence and retirement planning and Section 5 concludes.

2. Literature Review

The existing literature is analyzed as follows: (a) studies dealing with the impact of financial literacy on individual retirement planning; (b) studies on the financial literacy overconfidence and its economic effect.

2.1 Financial literacy and retirement planning

The first strand of literature dealing with financial literacy and retirement planning is well-established with a very large number of contributions. The most common measure of financial literacy is based on the surveys first proposed by Lusardi and Mitchell (2011) which was part of 2004 Health and Retirement Study (HRS). Lusardi and Mitchell (2011, 2007a) pioneer to find financial knowledge is directly related to retirement planning, those who display a higher level of financial knowledge are more likely to plan and to succeed in their planning. Using a new database of the Rand American Life Panel (ALP), Lusardi and Mitchell (2007b, 2017) confirm the positive relationship between knowledge of financial matters and retirement planning.

More recently, the issue of the relationship between financial literacy and retirement planning has also been explored in many countries other than the US. In these countries, financial literacy levels are widely found to be lower among the young, women and those with low income or low educational attainment, and a high financial literacy increases the

probability of having a retirement savings plan. This finding has been observed in Sweden, Italy, German, Japan (Almenberg and S  ve-S  derbergh, 2011; Fornero and Monticone, 2011; Bucher-Koenen and Lusardi, 2011; Sekita S, 2011). Still, in some countries like Finland, financial literacy measured by some extended questions in addition to the basic financial literacy questions positively affects the retirement planning, though the basic financial literacy does not (Kalmi and Ruuskanen, 2011).

2.2 Financial literacy overconfidence

The second strand of literature deals with the financial literacy overconfidence and its effect on financial decision behaviors. On the one hand, overconfidence has long been found to affect various investment decisions in the finance literature (Malmendier and Tate, 2005; Statmen et al., 2006; Barber and Odean, 2001; Grinblatt and Keloharju, 2009). For instance, Barber and Odean (2000, 2001) find that overconfident investors trade excessively and end up with lower returns. Grinblatt and Keloharju (2009) provide the evidence that overconfidence promotes tendency in stock market trades.

On the other hand, financial literacy overconfidence is a specific form of overconfidence, referring that individual's self-assessed financial literacy exceeds the actual level. In the limited existing literature, financial literacy overconfidence has been linked to some financial decision behaviors (Glaser and Weber, 2010). It is found that financial literacy overconfidence is positively correlated with stock market participation and under-confidence is negatively correlated to stock market participation (T. Xia et al., 2014; Chu Z, 2017). Moreover, Van Rooij et al. (2012) identify that the individuals whose subjective financial literacy is higher than objective one are more likely to be a serious planner for retirement.

3. Data and description

In 2018, China Center for Insurance and Risk Management of Tsinghua University and Aegon Center for Longevity and Retirement jointly conducted a nationwide online retirement readiness survey. Our analysis is based on this survey data of 1432 individuals aged between 25–65 from all provinces in China. The reason for restricting the respondents to be non-retired and aged between 25-65 is to exclude those in the decumulation or education phase of the life cycle.

3.1 Objective financial literacy measurement

On the basis of the financial literacy surveys and literature, we identified and adopted seven most representative questions that measure objective financial literacy. The famous ‘Big Three’ financial literacy questions, proposed by Lusardi and Mitchell (2011) in their analysis of older Americans in the 2004 Health and Retirement Study, are incorporated. The first two questions measure respondents’ ability to perform simple interest calculation and understand the effect of inflation. The third question evaluates the knowledge of risk diversification which is crucial to making informed investment decisions. To further evaluate the respondents’ financial knowledge, four additional questions were asked, covering fundamental concepts of mortgage repayment, bond pricing and simple calculation about income tax. The objective financial literacy (objective FL) is derived by adding together the scores of all seven questions, and has a theoretical range of 0–7. The precise wording of all objective financial literacy questions is in Table 1.

[Insert Table 1 here]

Summary statistics of responses to these questions appear in Table 2. Approximately 31.7% of respondents could correctly answer the Big Three questions. This number is close to that in the U.S. (35%) and Japan (26.9%). Specifically, 78% correctly answered the interest rate question, approximately 72% of respondents got the inflation question correct, and another 43% could answer the risk diversification question. In addition, among all the respondents, only 5.1% got all the seven questions right. These are discouragingly low numbers, in view of the complex financial decisions that individuals confront in the current economic environment.

[Insert Table 2 here]

3.2 Subjective financial literacy measurement

It is instructive to identify the mismatch between respondents’ perceived versus actual financial knowledge. In order to know how the respondents think of their own financial

knowledge, the survey also asks about subjective financial literacy. The respondents' subjective financial literacy (subjective FL) is measured by asking:

On a scale from 1 to 5, where 1 means very incapable and 5 means very capable, how would you assess your overall understanding of financial matters when you prepare for retirement?

Thus, subjective FL value varies within the range of 1–5.

It is found that although many respondents perform poorly in their responses to the objective financial literacy questions, as just described above, they tend to believe they do rather well. Around two-fifths (70%) award themselves the top knowledge scores (4 and 5), and only 4% give themselves failing marks (1–2), and the other 27% perceive themselves to be the medium level (3). That is, almost 70% of respondents believe they are above-median with regard to financial knowledge, which is in great contrast to the findings on the respondents' actual financial knowledge.

Figure 1 shows the distribution of subjective and objective financial literacy scores. The area of each circle is proportional to the number of subjects with that value. It is obvious that there are more people in the upper left area than in the bottom right part, indicating a higher percentage of overconfidence in our sample.

[Insert Figure 1 here]

3.3 Financial literacy overconfidence and under-confidence

The comparison between the results of objective and subjective financial literacy reveals that people tend to be overconfident about their financial literacy. It is meaningful to quantify and investigate this mismatch between perceived and actual financial knowledge.

To this end, consistent with T. Xia et al. (2014), we measure financial literacy overconfidence by the difference between the consumers' subjective and objective financial literacy scores. Due to different scales, the self-assessed and actual financial literacy are not directly comparable. Throughout our paper, the financial literacy overconfidence (FL overconfidence) refers to the phenomenon that some respondents perform poorly (below average) in their responses to the objective financial concepts but believe they do rather well

(above average). The opposite of overconfidence is the financial literacy under-confidence (FL under-confidence), referring to the group subjectively underestimate their financial literacy. Both variables are binary dummies taking the value of 1 if the respondent is overconfident or under-confident, respectively, and 0 otherwise.

In Figure 2, the mean of subjective (3.957) and objective scores (4.209) divide the graph into four regions, each represents a different type of observation. We are interested in respondents in Regions II and IV. The respondents in Region II are overconfident about their financial literacy, who has a below average objective FL but subjectively believe to possess an above average financial literacy. Conversely, the respondents in Region IV, being under-confident about their financial literacy, have higher objective FL scores but lower subjective FL scores.

[Insert Figure 2 here]

[Insert Table 3 here]

To determine which sub-groups are more confident about their financial knowledge, we compare the overconfidence of each group, as indicated in Table 3. The statistics show that the relationship between FL overconfidence and age is a U-shape curve. Those in the group aged 40-49 are most modest, those younger or older are more overconfident. Table 3 also shows that FL overconfidence is negatively correlated with educational attainment. The least financial literacy overconfident are those who lack a college degree; only about one-third of such respondents could answer the risk diversification question correctly. The percentage of FL overconfidence decreases with the education level, this number is reduced to 31.3% for the groups with postgraduate degrees (master & Ph.D.).

3.4 Retirement planning measurement

The main interest and dependent variable of this paper is retirement planning. Retirement planning is an inherently complex task since the planner needs to collect and process a lot of information. It is hard to measure the retirement planning level of an individual comprehensively with a single question.

There are some popular questions devoted to measuring the wellness of respondents'

retirement planning. One of the most widely-adopted questions is: “*How much have you thought about retirement: A lot, some, little, or hardly at all?*”. This question has been developed and adopted by the 2004 HRS and Lusardi and Mitchell (2011a, 2017). Another question measuring how well respondents have prepared their retirement plan is “*Do you have a savings plan for the future? (i) I have a specific plan, (ii) I have a rough plan, (iii) I do not have a plan now, but I am going to make a plan in the future, (iv) I do not have a plan now, and I am not going to make a plan in the future.*” (Sekita S., 2011). This question is largely similar to the previous one, but less subjectively dependent. The perception on savings and investments for retirement is another indicator of retirement planning readiness, the typical question is: “*Have you ever tried to figure out how much you would need to save for retirement?*”. This question is also widely adopted in the literature (Alessie R et al, 2011; Lusardi and Mitchell, 2011a).

It is noteworthy that, the respondents’ response to the above-mentioned retirement planning readiness questions are based on their subjective assessment of retirement planning to some extent. A potential concern is that an overconfident individual may overstate both financial literacy and wellness of planning. Thus, the finding of some previous literature that the FL overconfidence is positively related to individual’s retirement planning remains to be confirmed with more objective retirement readiness measurement. Therefore, we aim to confirm and investigate how FL overconfidence affects individual’s actual behaviors of retirement preparation, like purchasing a variable annuity. To that end, our survey contains a series of questions on retirement planning that ask both the respondents’ self-perception of retirement preparation and whether they have actual behaviors including joining private pension or purchasing a variable annuity. The exact wording of the retirement planning questions is in Table 4.

[\[Insert Table 4 here\]](#)

Therefore, the four questions in the survey (Table 4) test the respondent’s retirement planning from perspectives of *knowing*, *planning* and *real actions*. The traditional questions 1 and 2 are based on the respondent’s perception and self-assessment to some extent, which

measure the *planning* and *knowing* aspects of retirement planning. Questions 3 and 4 supplement the measurement by asking *real actions* related to retirement preparation.

The responses to the retirement planning questions appear in Table 4. It shows that around 76% of respondents have a plan for their retirement, but only approximate 13% sub-group have written down their plan. The percentage of respondents who do not have any plan for retirement is around 24%. Most people believe they have a good idea of the total value of their retirement assets (level 4 & 5), and only less than 30% of respondents think they are not well aware of the savings and investments for retirement.

As for the actual actions taken by the respondents to prepare for retirement, around 44% of the respondents have participated in a private pension, which is explained as pension sources other than the national public pension. And it is not surprising to find that only 8% have purchased a variable annuity as the market of variable annuity is rather limited due to its high investment threshold. Hence, from the simple observation we can draw the conclusion that there are more planners who have drafted a plan than those who have turned their retirement plan into real actions.

4. Empirical analysis

4.1 Methodology

In this paper, we aim at determining the effects of the financial literacy bias. The survey question considers and measures both the self-assessment and effective behaviors of respondents' retirement planning. The mismatch between the respondent's perceived and actual financial knowledge, or the FL confidence level, is captured by the introduced FL overconfidence and under-confidence.

Before regression analysis, we expect that the individuals' confidence in financial knowledge might assert an independent effect on their retirement planning behaviors. Individuals who are overly confident about their knowledge might purchase products that they do not fully understand and end up making more actual preparation for retirement. In addition, the over-confident group might also be optimistic about the wellness of their retirement planning. We perform a multivariate analysis of the relationship between retirement planning and financial literacy. The main regression model is as follows:

$$\begin{aligned} \text{Retirement planning} = & \alpha + \beta_1 \text{ FL overconfidence} + \beta_2 \text{ FL under-confidence} \\ & + \beta_3 \text{ Objective FL} + \gamma \cdot \text{control variables} + \varepsilon. \end{aligned} \quad (1)$$

The dependent variables are variables indicating the readiness of retirement planning, including all variables from questions in Table 4. These four dependent variables in this paper can be classified into two categories: one measures the respondent's retirement planning assessment, and the other measures the retirement planning actions.

Ordinary least squares (OLS) regressions are applied to the former two self-assessment retirement planning variables. A probit regression was used for the latter two because the dependent variables are binary. The list of controls includes objective financial literacy, age, gender and marital status; other controls include income levels, region of residence, and educational attainment. Table 5 and Table 6 provide the definition and summary statistics of the regression variables, respectively.

[Insert Table 5 here]

[Insert Table 6 here]

4.2 Regression result

In our survey, the respondent's retirement planning is measured by their assessment of retirement plan and actual financial actions taken for preparation. Table 7 reports the results of ordinary least squares (OLS) regressions, the dependent variable is respondent's retirement planning assessment, based on the question 1 & 2 of Table 4. In Table 8, we report the result of probit regressions for the binary dependent variables, which indicate the planner's financial actions in preparing for retirement.

Consistent with the prior literature, the coefficients of the objective financial literacy in both Table 7 and Table 8 are significantly positive. Thus, those who answered more questions correctly are more likely to plan for retirement and also take actual preparation actions.

[Insert Table 7 here]

[Insert Table 8 here]

The effect of FL overconfidence and under-confidence are of our main interest. In both Table 7 and 8, FL overconfidence has an independent and significantly positive effect to promote retirement preparation before and after controlling for financial literacy. Table 7 shows that FL overconfidence independently and significantly promotes both individual's preparation of the readiness of retirement planning and awareness of retirement savings. From columns 1, 2 and 3, we see that FL overconfidence significantly promotes the individual's retirement plan preparation with or without controlling financial literacy. The similar results in columns 4 to 6 also suggest that those who are overconfident about their financial knowledge are more likely to be aware of the value their retirement savings and investments.

Moreover, the probit models in Table 8 confirm the finding that FL overconfidence could explain actual retirement preparation behaviors, after controlling some variables including financial literacy. This finding indicates that individual's high-confidence about their financial knowledge motivates them not only to prepare more for making a retirement plan, but also to put the retirement plan into effect. Thus, the overconfident group are more likely to join private pension and purchase a variable annuity.

On the other hand, the coefficients of FL under-confidence of the regression are significantly negative, which indicates that individual's pessimistic belief about their financial literacy decrease their motivation and actions in retirement planning. Besides, the coefficients of FL under-confidence in columns 3 and 6 of both Table 7 and 8 are significantly negative, without affecting the significant effects of financial literacy and FL overconfidence. This independent effect might be intuitively explained by that individuals who are overly modest about their knowledge refrain from drafting retirement plan and taking actions like buying financial products due to their under-confidence.

Apart from the financial literacy and its self-assessed bias, some other variables are also found to affect retirement planning. First, consistent with many previous literature, the personal income positively encourages the retirement planning and actions taking according to our result. Second, while men are not better at making retirement plan than women, they are more aware of their financial status with respect to retirement. Third, in some regressions, the education and age are found to promote retirement preparation.

5. Conclusion

Understanding the role of financial literacy on individual's retirement planning, including both the subjective and objective financial literacy, is important in this aging era. As individuals are taking more responsibility for securing their own financial well-being in retirement around the world, it is critical to understand what types of financial education programs are most effective.

Our study contributes to the limited existing literature on the role of subjective financial literacy and financial literacy bias in retirement planning and preparations. The findings in this paper suggest that in addition to individual's actual financial literacy, their optimistic or pessimistic beliefs towards their financial literacy also affect the retirement planning behaviors. In particular, our empirical results show that respondents with more confidence in their financial knowledge have a higher propensity to plan and take actions for retirement. On the other hand, individuals who are under-confident are less likely to have a better retirement preparation and actual actions. Thus, our work confirms the effective role of subjective financial literacy and financial literacy bias in retirement planning with data from China.

Another contribution is that this paper extends current research on the effects of financial literacy confidence on retirement planning by taking account of the planner's actual actions for retirement. We have shown that overconfident individuals are more likely to not only have thought about their retirement plan but also take actual actions to prepare for retirement. From our result, we can see that the propensity of participating in a private pension or purchasing a variable annuity of overconfident respondents is greater than the respondents who have normal or lower confidence in their financial literacy. Hence, our result suggests that individual's overconfidence in their financial literacy is good for increasing the likelihood and motivation for retirement planning.

Therefore, this finding is insightful for providing education program for retirement planning. Through this study, we have shown that the individuals' overconfidence towards their financial ability encourages a better planning. Hence, planners do not only need to have high financial knowledge, but also they need to have confidence in their level of knowledge to motivate better retirement planning and actual actions. From the perspective of enhancing

retirement planning, efforts are needed to encourage planners that the financial knowledge for retirement planning is obtainable and they are able to take a more active role in financial activities for retirement planning.

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Table 1 Objective financial literacy questions

Questions	Answers
1. Suppose you have 100 yuan in a savings account and the interest rate was 2% per year. After 5 years, how much do you think you would have in the account if you leave the money to grow?	(i) More than 102 yuan (ii) Exactly 102 yuan (iii) Less than 102 yuan (iv) Do not know
2. Imagine that the interest rate on your savings account is 1% per year and inflation is 2% per year. After 1 year, how much will you be able to buy with the money in this account?	(i) More than today (ii) Exactly the same (iii) Less than today (iv) Do not know
3. Please tell me whether this statement is true or false. 'Buying a single company's stock usually provides a safer return than a stock mutual fund'.	(i) True (ii) False (iii) Do not know
4. Suppose you take a mortgage loan from the bank. One repayment term option would require you to repay the mortgage over 15 years and the other option is to repay the mortgage over 30 years. Both loans have the same interest rate (4%). Then the monthly mortgage repayment of the 15-year loan is higher than the 30-year loan.	(i) This statement is right (ii) The statement is wrong (iii) I don't know
5. Imagine that you want to take out a mortgage loan from your bank today. You anticipate that interest rates are set to rise over the next few months. Which type of mortgage loan would you choose to take out?	(i) Fixed rate loan (ii) Floating rate loan (iii) I don't know
6. Imagine that you buy a bond today for a price of 97 yuan. In a year's time when the bond reaches maturity it is expected to be worth 100 yuan. If interest rates rise tomorrow, then how will the market price of this bond change?	(i) Its price will rise (ii) Its price will drop (iii) Its price will remain unchanged (iv) I don't know
7. Imagine that your income puts you in the 25% tax bracket. You earn 5,000 yuan per month and contribute 1,000 yuan before tax to cover all of your social security and pension plan payments. What would your take-home pay be?	(i) 5000 yuan (ii) 4000 yuan (iii) 3000 yuan (iv) 2000 yuan (v) I don't know

Table 2 Summary statistics on objective financial literacy questions (%)

Question	Correct responses(%)	Wrong responses(%)	Do not know(%)
Q1: Compound interest	78.1	17.7	4.2
Q2: Inflation	71.9	23.0	5.1
Q3: Risk diversification	43.3	27.4	29.3
Q4: Mortgage repayment	67.3	21.6	11.1
Q5: Choosing a mortgage loan	70.0	24.2	5.8
Q6: Bond pricing	28.4	60.4	11.2
Q7: Income tax	58.2	35.0	6.8

Table 3 Distribution of financial literacy overconfidence and under-confidence by age, sex,

education

	Financial literacy overconfidence(%)	Financial literacy under-confidence(%)
<i>Age</i>		
25-29	41.8	10.4
30-34	43.2	12.8
35-39	44.3	16.9
40-44	32.4	21.7
45-49	38.9	14.5
50-54	45.8	14.2
55-59	45.0	8.3
60-64	56.5	8.7
<i>Sex</i>		
Male	38.0	16.2
Female	45.5	12.1
<i>Education</i>		
<College	56.6	10.1
College graduate	40.6	14.4
Post graduate	31.9	17.2

Table 4 Retirement planning questions

Questions	Answers (%)
1. Which of the following best describes your retirement planning? (Retirement plan)	(i) I do not have a plan (24.16%) (ii) I have a plan, but it is not written down (62.78%) (iii) I have a written plan (13.06%)
2. To what extent (1-5) do you agree with the statement: I have a very good idea of the total value of all my personal retirement savings and investments? (Retirement assets awareness)	(i) Strongly disagree (0.75%) (ii) Somewhat disagree (5.30%) (iii) Neither agree or disagree (22.12%) (iv) Somewhat agree (49.05%) (v) Strongly agree (22.78%)
3. Are you currently participating in a private pension to prepare for your retirement? (Private pension)	(i) Yes (43.79%) (ii) No (56.21%)
4. Are you currently having a variable annuity to prepare for your retirement? (Variable annuity)	(i) Yes (91.88%) (ii) No (8.12%)

Notes: The private pension here refers to pension income source other than the public pension, including the DC/DB pension and commercial pension product purchased from insurers.

Table 5 Variables and definition

Variable	Variables description
Retirement plan	Readiness of retirement planning (range 1-3)
Retirement assets awareness	Awareness of total value of retirement savings and investments (range 1-5)
Private pension	Participation in a private pension to prepare for retirement, 1 indicating yes and 0 indicating no
Variable annuity	Having a variable annuity to prepare for retirement, 1 indicating yes and 0 indicating no
FL overconfidence	For respondents with <i>objective FL</i> < 4.209 and <i>subjective FL</i> > 3.957, <i>overconfidence</i> = 1, otherwise = 0
FL under-confidence	For Respondents with <i>objective FL</i> > 4.209 and <i>subjective FL</i> < 3.957, <i>under-confidence</i> = 1, otherwise = 0
Objective FL	The objective financial literacy score (range 0-7)
Subjective FL	The subjective financial literacy score (range 1-5)
Gender	1 for male, 0 for female
Personal income	Annual income (the amount is categorized into range 1-9)
Children	Number of children
Marriage	Marital status, 1 means married, and 0 otherwise
Age	Age (range 25-65)
<College	Education below college = 1, otherwise = 0 (benchmark)
College graduate	College diploma = 1, otherwise = 0
Post graduate	Education above college = 1, otherwise = 0
Risk aversion dummy 1 (low)	Degree of risk averse (range 1-4)
Risk aversion dummy 2	Degree of risk averse (range 1-4)
Risk aversion dummy 3	Degree of risk averse (range 1-4)
Risk aversion dummy 4 (high)	Degree of risk averse (range 1-4)

Table 6 Summary statistics

Variable	Mean	Std. Dev.	Min	Max
Retirement plan	1.889	0.600	1	3
Retirement assets awareness	3.878	0.845	1	5
Private pension	0.573	0.495	0	1
Variable annuity	0.083	0.276	0	1
FL overconfidence	0.417	0.493	0	1
FL under-confidence	0.142	0.349	0	1
Objective FL	4.209	1.585	0	7
Subjective FL	3.957	0.804	1	5
Gender	0.514	0.500	0	1
Personal income	7.165	2.173	1	9
Children	0.969	0.544	0	4
Marriage	0.878	0.327	0	1
Age	41.675	10.034	25	64
<College	0.132	0.339	0	1
College graduate	0.754	0.431	0	1
Post graduate	0.114	0.318	0	1
Risk aversion dummy 1 (low)	0.359	0.480	0	1
Risk aversion dummy 2	0.286	0.452	0	1
Risk aversion dummy 3	0.134	0.341	0	1
Risk aversion dummy 4 (high)	0.153	0.360	0	1

Table 7 Multivariate analysis of retirement planning assessment: OLS regression result

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)
	Retirement plan			Retirement assets awareness		
FL over-confidence	0.248*** (0.035)	0.304*** (0.042)	0.263*** (0.042)	0.272*** (0.058)	0.573*** (0.067)	0.508*** (0.067)
FL under-confidence			-0.311*** (0.045)			-0.508*** (0.086)
Objective FL		0.028** (0.014)	0.046*** (0.014)		0.154*** (0.022)	0.179*** (0.022)
Gender	0.021 (0.033)	0.017 (0.033)	0.021 (0.032)	0.128** (0.053)	0.116** (0.053)	0.124** (0.052)
Personal income	0.073*** (0.008)	0.072*** (0.008)	0.071*** (0.008)	0.049*** (0.014)	0.045*** (0.014)	0.046*** (0.013)
Children	0.042 (0.036)	0.041 (0.035)	0.041 (0.035)	-0.004 (0.056)	-0.001 (0.054)	-0.014 (0.054)
Marriage	-0.009 (0.057)	-0.010 (0.056)	-0.020 (0.054)	0.114 (0.093)	0.110 (0.091)	0.096 (0.091)
Age	-0.000 (0.002)	-0.001 (0.002)	-0.001 (0.002)	0.010*** (0.003)	0.007** (0.003)	0.007** (0.003)
College graduate	-0.057 (0.055)	-0.065 (0.054)	-0.067 (0.054)	0.194** (0.096)	0.157* (0.093)	0.148 (0.094)
Post graduate	0.018 (0.073)	0.005 (0.073)	0.005 (0.073)	0.260** (0.123)	0.198* (0.120)	0.187 (0.120)
Risk aversion dummy 1 (low)	0.193*** (0.068)	0.155** (0.068)	0.121* (0.067)	0.235** (0.117)	0.022 (0.118)	-0.023 (0.113)
Risk aversion dummy 2	0.279*** (0.070)	0.249*** (0.070)	0.221*** (0.069)	0.217* (0.118)	0.052 (0.118)	0.019 (0.113)
Risk aversion dummy 3	0.114 (0.075)	0.082 (0.075)	0.052 (0.074)	0.155 (0.130)	-0.024 (0.129)	-0.062 (0.124)
Risk aversion dummy 4 (high)	0.142* (0.075)	0.100 (0.075)	0.078 (0.075)	0.261** (0.132)	0.026 (0.134)	0.000 (0.129)
Constant	0.898*** (0.159)	0.826*** (0.163)	0.871*** (0.166)	2.628*** (0.304)	2.184*** (0.309)	2.278*** (0.284)
Region Effect	√	√	√	√	√	√
Observations	1,266	1,266	1,266	1,063	1,063	1,063
R-squared	0.180	0.183	0.209	0.100	0.140	0.171

Standard error in parentheses. Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1.

Table 8 Multivariate analysis of retirement planning actions: Probit regression result

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)
	Private pension			Variable annuity		
FL over-confidence	0.149*	0.309***	0.281***	0.380***	0.632***	0.644***
	(0.078)	(0.100)	(0.101)	(0.113)	(0.149)	(0.152)
FL under-confidence			-0.204*			0.065
			(0.114)			(0.175)
Objective FL		0.081**	0.092***		0.119**	0.117**
		(0.032)	(0.033)		(0.053)	(0.053)
Gender	0.025	0.013	0.016	-0.028	-0.031	-0.033
	(0.075)	(0.075)	(0.075)	(0.109)	(0.110)	(0.110)
Personal income	0.034*	0.033*	0.032*	0.046	0.044	0.044
	(0.019)	(0.019)	(0.019)	(0.030)	(0.031)	(0.031)
Children	0.072	0.068	0.069	0.325***	0.325***	0.324***
	(0.078)	(0.077)	(0.077)	(0.103)	(0.105)	(0.105)
Marriage	0.080	0.077	0.070	-0.352*	-0.348*	-0.347*
	(0.129)	(0.128)	(0.128)	(0.201)	(0.200)	(0.201)
Age	0.001	-0.001	-0.001	0.016***	0.014**	0.014**
	(0.004)	(0.004)	(0.004)	(0.006)	(0.006)	(0.006)
College graduate	-0.001	-0.027	-0.027	0.001	-0.033	-0.032
	(0.121)	(0.121)	(0.121)	(0.163)	(0.167)	(0.166)
Post graduate	0.085	0.048	0.050	0.432*	0.380*	0.383*
	(0.162)	(0.163)	(0.163)	(0.222)	(0.225)	(0.224)
Risk aversion dummy 1 (low)	0.622***	0.522***	0.501***	0.096	-0.019	-0.011
	(0.157)	(0.162)	(0.163)	(0.257)	(0.269)	(0.266)
Risk aversion dummy 2	0.533***	0.458***	0.441***	0.430*	0.337	0.343
	(0.160)	(0.163)	(0.164)	(0.253)	(0.259)	(0.257)
Risk aversion dummy 3	0.545***	0.465**	0.446**	0.457*	0.349	0.356
	(0.177)	(0.181)	(0.181)	(0.271)	(0.277)	(0.274)
Risk aversion dummy 4 (high)	0.597***	0.484***	0.471***	0.187	0.056	0.062
	(0.173)	(0.179)	(0.180)	(0.276)	(0.289)	(0.286)
Constant	-0.875**	-1.085***	-1.061***	-2.768***	-3.148***	-3.163***
	(0.378)	(0.391)	(0.390)	(0.498)	(0.526)	(0.526)
Region Effect	√	√	√	√	√	√
Observations	1,263	1,263	1,263	1,232	1,232	1,232
Pseudo R-squared	0.0514	0.0551	0.057	0.1	0.107	0.1071

Standard error in parentheses. Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1.

Figure 1 Distribution of subjective and objective financial literacy scores

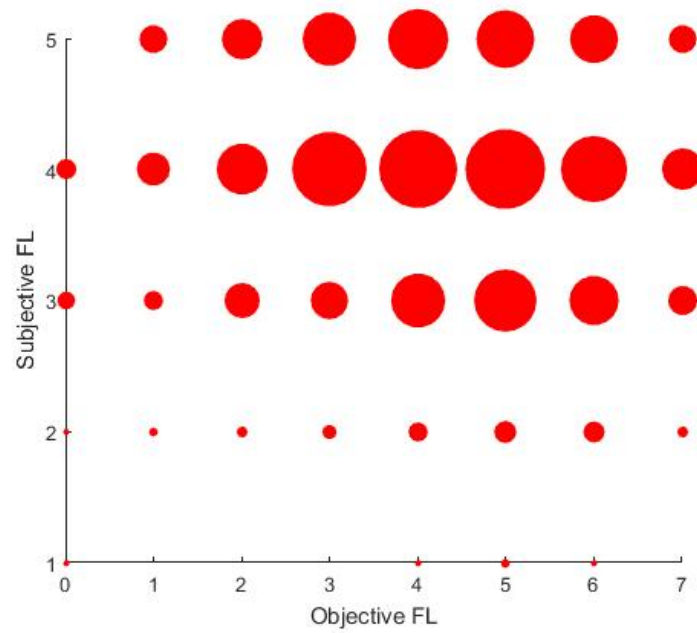


Figure 2 Financial literacy confidence types

